

BRST Symmetry:

Start with original Polyv. Action

$$S^{\text{orig.}} = -\frac{1}{4\alpha'} \int d^2\zeta \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$$

• This action has the following symmetries

(i) Reparametrization:

$$\begin{aligned}\delta g_{\alpha\beta} &= \nabla_\alpha V_\beta + \nabla_\beta V_\alpha \\ \delta X^\mu &= V^\alpha \partial_\alpha X^\mu\end{aligned}$$

→ μ : is inert under a general coord transf.

$$g^{\alpha\beta} = -(\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)$$

$$\delta g = g g^{\rho\sigma} \delta g_{\rho\sigma} = 2g \nabla_\alpha V^\alpha$$

(ii) Weyl invariance:

$$\begin{cases} \delta g_{\alpha\beta} = c^\phi g_{\alpha\beta} \\ \delta X^\mu = 0 \end{cases}$$

* We now Gauge-fix the above action by introducing Ghost fields: (b, c) . This will leave the prop. invariant.

→

BRST PRIMER

$$A^x B_x \equiv \int dx A(x) B(x)$$

• Gauge symmetry: $\delta \Phi^F = \epsilon^\alpha \delta_\alpha \Phi^F$
 $\epsilon^\alpha = \epsilon^\alpha(x)$

• Invariance of this Gauge symmetry
 $S[\Phi^F + \delta \Phi^F] = S[\Phi^F] \Rightarrow \delta S[\Phi^F] = 0$

• Gauge-fixing condition: $\boxed{I^A(\Phi^F) = 0}$

Rules:

(i) Introduce Ghost fields:

$$e^a \longrightarrow c^a \quad \forall a$$

c^a has opposite statistics to e^a

ex: If $e^a \equiv (\lambda^1, \lambda^2)$ $\lambda^a \lambda^b = \lambda^b \lambda^a$

Then $c^a = (c^1, c^2)$ $c^a c^b = -c^b c^a$

(ii) Introduce Anti-Ghost fields:

- b_a as well as Auxiliary fields B_A , one for each Gauge fixing condition $F^A(\Phi^I) = 0$

$$\left. \begin{array}{l} b_A \longleftrightarrow c_A \\ B_A \longleftrightarrow G_A \end{array} \right\} \begin{array}{l} \text{Same} \\ \text{Statistics} \end{array}$$

(iii) We add the following terms to S^{orig} :

(a) $S^{\text{gf}} = -i B_A F^A(\Phi^I)$

[Gauge Fixing]

(b) $S^{\text{gh}} = b_A c^a \partial_a F^A$

[Ghost Action]

$$S = S^{\text{or}} + S^{\text{gh}} + S^{\text{gf}}$$

Back to the String:

* Recall that we may locally find a $f^a \phi(\xi)$
 $\phi: \Sigma \rightarrow \mathbb{R}$, s.t. $g_{\alpha\beta}$ is locally conformally equivalent to $\eta_{\alpha\beta}$

$$g_{\alpha\beta} = e^{\phi(\xi)} \eta_{\alpha\beta}$$

→ To rid ourselves of ϕ is our charges we look to Gauge fix the Weyl invariant term

$$\sqrt{-g} g^{\alpha\beta} = \sqrt{-g'} g'^{\alpha\beta}$$

$$A = (\alpha\beta)$$

$$F^A[g] = 0 \equiv \sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta} = 0$$

(Gauge Fixing Condition)

$$F_A(g) = F_B(g) = 0$$

- From the BRST procedure this implies we introduce the action for Gauge fixing

$$\begin{aligned} S^{gf} &= -i B_{\alpha\beta} F^{\alpha\beta}(\hat{g}, \eta) \\ &= -i \int d^2\hat{\xi} B_{\alpha\beta} (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta}) \end{aligned}$$

→ We normalize this as

$$S^{gf} = -\frac{i}{\pi} \int d^2\hat{\xi} \lambda_{\alpha\beta} (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})$$

Where $B_{\alpha\beta} i = \lambda_{\alpha\beta}/\pi$

$\lambda_{\alpha\beta}$: (Auxiliary) Lagrange field

- * The Constraint $F_{\alpha\beta} = 0$ contains only two conditions since we are choosing

$$\sqrt{-g} g_{\alpha\beta} = \eta_{\alpha\beta} \quad \text{and so}$$

$$\sqrt{-g} \begin{pmatrix} g_{00} & 0 \\ 0 & g_{11} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{cases} \sqrt{-g} g_{00} = -1 \\ \sqrt{-g} g_{11} = 1 \end{cases}$$

- Only two components of $\lambda_{\alpha\beta}$ contribute to the Action.

→ $\lambda_{\alpha\beta} = \lambda_{\beta\alpha}$ and $\text{tr}(\lambda_{\alpha\beta}) = 0$ with respect to either $g^{\alpha\beta}$ or $\hat{g}^{\alpha\beta}$.

- * For Diffeomorphism invariance:

$$\delta x^\mu = U^\alpha \partial_\alpha x^\mu$$

$1/\alpha \leftrightarrow \epsilon^\alpha$ This is the parameter

$V^\alpha \mapsto \Lambda C^\alpha$ is the parameter.

$$V^\alpha \mapsto \Lambda C^\alpha$$

Thus,

$$\delta X^\mu = \Lambda C^\alpha \partial_\alpha X^\mu$$

$$\delta g_{\alpha\beta} = \nabla_\alpha (\Lambda C_\beta) + \nabla_\beta (\Lambda C_\alpha)$$

$$\Rightarrow \delta X^\mu = \Lambda C^\alpha \partial_\alpha X^\mu$$

$$\delta g^{\alpha\beta} = -\Lambda (\nabla^\alpha C^\beta + \nabla^\beta C^\alpha)$$

* The variation of all the fields

$$\delta_B b_{\alpha\beta} = \Lambda \lambda_{\alpha\beta}$$

$$\delta_B \lambda_{\alpha\beta} = 0$$

$$\delta_B C^a = -\frac{1}{2} K f_{\gamma\delta}^a C^\gamma C^\delta$$

From these

$$\delta_B b_{\alpha\beta} = \Lambda \lambda_{\alpha\beta}$$

$$\delta_B \lambda_{\alpha\beta} = 0$$

(other ghost fields)
BRST

• This implies: $b_{\alpha\beta} = -b_{\beta\alpha}$, $\text{tr}(b) = 0$

We take: $\delta_B C^a = \Lambda C^\gamma \nabla_\gamma C^a = \Lambda C^\gamma \partial_\gamma C^a$

$$[\delta_1, \delta_2] X^\mu = [E_1^\alpha \partial_\alpha, E_2^\beta \partial_\beta] X^\mu$$

$$= E_1^\alpha \partial_\alpha E_2^\beta - (1 \leftrightarrow 2)$$

$$= (E_1^\alpha \nabla_\alpha E_2^\beta - (1 \leftrightarrow 2)) X^\mu$$

$$\bullet E_1^\alpha \rightarrow \Lambda C^\alpha$$

$$E_2^\beta \rightarrow C^\beta$$

$$[\delta_1, \delta_2] X^\mu = \Lambda C^\alpha \nabla_\alpha C^\beta - C^\alpha \nabla_\alpha (\Lambda C^\beta)$$

$$= (2 \Lambda C^\alpha \nabla_\alpha C^\beta)$$

• Fields δC^a

* Note: BRST is nilpotent:

$$\begin{aligned}\delta_1 \delta_2 x^\mu &= \delta_1 (\Delta c^a \partial_a x^\mu) \\ &= \Delta_2 \delta c^a \partial_a x^\mu + \Delta c^a \partial_a (\delta_1 x^\mu) \\ &= \Delta_2 (\Delta_1 c^\gamma \partial_\gamma c^a) \partial_a x^\mu \\ &\quad + \Delta_2 c^a \partial_a (\Delta_1 c^a \partial_a x^\mu)\end{aligned}$$

Now add the Ghost Action:

$$S^{\text{gh}} = b_{\alpha\beta} c^\gamma \underbrace{\delta_\gamma F^{\alpha\beta}}_{\leftarrow \text{Int. over } \int d^2\xi}$$

$$= b_{\alpha\beta} c^\gamma \delta_\gamma (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})$$

$$S^{\text{gh}} = \int d^2\xi \underbrace{b_{\alpha\beta} c^\gamma}_{\leftarrow \text{Int. over } \int d^2\xi} [\delta_\gamma (\sqrt{-g}) g^{\alpha\beta} + \sqrt{-g} \delta_\gamma g^{\alpha\beta}]$$

$$\text{(i)} \quad \delta_\gamma (\sqrt{-g}) = -\frac{1}{2} \frac{1}{\sqrt{-g}} \delta_\gamma g = -\frac{1}{\sqrt{-g}} g \nabla_\delta (\Delta c^\delta)$$

$$\delta_\gamma (\sqrt{-g}) = \sqrt{-g} \Delta \nabla_\delta c^\delta$$

$$\begin{aligned}\text{(ii)} \quad \delta g^{\alpha\beta} &= -(\nabla^\alpha (\Delta c^\beta) + \nabla^\beta (\Delta c^\alpha)) \\ &= -\Delta (\nabla^\alpha c^\beta + \nabla^\beta c^\alpha)\end{aligned}$$

* Derivation of Ghost Action:

• We have

$$S_3 = b_{\alpha\beta} c^\gamma \delta_\gamma F^{\alpha\beta} (\mathbb{F})$$

Note: For EACH index summation we integrate over different coords

$$b_{\alpha\beta} c^\gamma \delta_\gamma F^{\alpha\beta} = \int d^2\xi' d^2\xi b_{\alpha\beta}(\xi') c^\gamma(\xi) \delta_\gamma(\xi) F^{\alpha\beta}(\xi')$$

$$\begin{aligned}&= \int d^2\xi d^2\xi' b_{\alpha\beta}(\xi') c^\gamma(\xi) \delta_\gamma(\xi) (\sqrt{-g} g^{\alpha\beta}(\xi') - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta}(\xi')) \\ &= \int d^2\xi d^2\xi' b_{\alpha\beta} c^\gamma \left[-\frac{1}{2} \frac{1}{\sqrt{-g}} \delta_\gamma g g^{\alpha\beta} + \sqrt{-g} \delta_\gamma g^{\alpha\beta} \right]\end{aligned}$$

* Must understand $\delta_\gamma g$ etc...

Note
 $\delta g^{\alpha\beta} = -1 (\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)$
 but $\delta g^{\alpha\beta} = \psi^\rho \delta_\rho g^{\alpha\beta}$, so using

$$\delta g^{\alpha\beta} = - (\nabla^\alpha V^\beta + \nabla^\beta V^\alpha) = \int d^2\xi' V^\rho(\xi') \delta_\rho(\xi) g^{\alpha\beta}$$

It's easy to see LHS:

$$-\int d^2\xi' (\nabla^\alpha V^\beta + \nabla^\beta V^\alpha)(\xi') \delta^2(\xi - \xi')$$

$$= \int d^2\xi' (V^\beta \nabla^\alpha \delta^2(\cdot) + V^\alpha \nabla^\beta \delta^2(\cdot))$$

$$= \int d^2\xi' (\delta_\rho^\beta V^\rho \nabla^\alpha \delta^2 + \delta_\rho^\alpha V^\rho \nabla^\beta \delta^2)$$

$$= \int d^2\xi V^\rho (\delta_\rho^\beta \nabla^\alpha \delta^2 + \delta_\rho^\alpha \nabla^\beta \delta^2)$$

We conclude

$$(\delta_\rho g^{\alpha\beta})(\xi) = \delta_\rho^\beta \nabla^\alpha \delta^2(\xi - \xi') + \delta_\rho^\alpha \nabla^\beta \delta^2(\xi - \xi')$$

Since

$$\delta g^{\alpha\beta} = -g^{\alpha\gamma} \delta g_{\gamma\delta} g^{\delta\beta} \text{ we also have}$$

(as in the set of notes)

$$-g^{\alpha\gamma} \delta g_{\gamma\lambda} g^{\lambda\beta} = [\delta_\rho^\beta (\nabla^\alpha \delta^2) + \delta_\rho^\alpha (\nabla^\beta \delta^2)]$$

$$(\delta_{\hat{\alpha}}^\alpha \delta_\rho g_{\gamma\lambda} \delta_{\hat{\beta}}^\beta) = - (g_{\alpha\hat{\alpha}} g_{\beta\hat{\beta}}) [\quad]$$

$$\delta_\rho g_{\hat{\alpha}\hat{\beta}} = - [g_{\hat{\beta}\rho} (\nabla_{\hat{\alpha}} \delta^2) + g_{\rho\hat{\alpha}} (\nabla_{\hat{\beta}} \delta^2)]$$

$\hat{\alpha} \rightarrow \alpha, \hat{\beta} \rightarrow \beta$: we conclude:

$$\delta_\rho g_{\alpha\beta}(\xi) = -g_{\alpha\rho}(\xi) (\nabla_\beta \delta^2(\xi - \xi')) - g_{\beta\rho} \nabla_\alpha \delta^2(\xi - \xi')$$

* If we now gather all of the above we find that:

$$b_A c^\lambda \delta_\lambda F^A = \int d^2 \xi d^2 \xi' b_{\alpha\beta}(\xi) c^\lambda(\xi') \delta_\lambda(\xi') F^{\alpha\beta}$$

$$= \int (d^2 \xi d^2 \xi') b_{\alpha\beta}(\xi) c^\lambda(\xi') \left[\delta_\lambda g(-\frac{1}{2}) \frac{1}{\sqrt{-g}} + \sqrt{-g} \delta g^{\alpha\beta} \right]$$

$$\rightarrow [g g^{\gamma\delta} \delta_\lambda g_{\gamma\delta} g^{\alpha\beta} (-\frac{1}{2}) \frac{1}{\sqrt{-g}} + \sqrt{-g} \delta_\lambda g^{\alpha\beta}]$$

$$= [(\sqrt{-g}/2) g^{\gamma\delta} \delta_\lambda g_{\gamma\delta} + \sqrt{-g} (-g^{\alpha\gamma} \delta_\lambda g_{\gamma\delta} g^{\beta\delta})]$$

$$= (\sqrt{-g}) (g^{\gamma\delta} g^{\alpha\beta} / 2 - g^{\alpha\gamma} g^{\beta\delta}) \delta_\lambda g_{\gamma\delta}$$

$$= \sqrt{-g} [\frac{1}{2} g^{\gamma\delta} g^{\alpha\beta} - g^{\alpha\gamma} g^{\beta\delta}] (-g_{\gamma\lambda} \nabla_\delta \delta^2 - g_{\delta\lambda} \nabla_\gamma \delta^2)$$

Plugging this back in:

$$\int d^2 \xi d^2 \xi' b_{\alpha\beta} c^\lambda (\sqrt{-g}) (\frac{1}{2} g^{\gamma\delta} g^{\alpha\beta} - g^{\alpha\gamma} g^{\beta\delta}) \delta g_{\gamma\delta}$$

- Note we required $\text{tr}(b) = 0 = g^{\alpha\beta} b_{\alpha\beta}$ so the first term vanishes, bruh.

$$\int d^2 \xi d^2 \xi' b^{\gamma\delta}(\xi) c'^\lambda \sqrt{-g} (g_{\gamma\lambda} \nabla_\delta \delta^2 + g_{\delta\lambda} \nabla_\gamma \delta^2)$$

$$\nabla_\delta = \nabla_\delta(\xi')$$

→ The final term $-g^{\alpha\beta} \nabla_\delta c^\delta$ is not present: ...
Not sure where this comes from...

$$S^{\text{gh}} = -\frac{1}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (\nabla^\alpha c^\beta + \nabla^\beta c^\alpha - g^{\alpha\beta} \nabla_\delta c^\delta)$$

oh...

the final term is the trace term which I got rid of... Not sure why he's keeping it if we had $\text{tr}(b) = 0$... okay.

*Quicker Method:

$$\Lambda(S^g h + S^g f) = -\frac{1}{\pi} \delta \int d^2 \xi \, b_{\alpha\beta} (\sqrt{-g} g^{\alpha\beta} - \sqrt{-\hat{g}} \hat{g}^{\alpha\beta})$$

$$= -\frac{1}{\pi} \delta_B \int d^2 \xi \, b_{\alpha\beta} F^{\alpha\beta}$$

and since $\delta^2 = 0$

$$\delta[\Lambda(S^g h + S^g f)] = 0$$

$$* (c^a)^\dagger = c^a \Leftrightarrow b_{\alpha\beta}^\dagger = -b_{\alpha\beta}$$

$$\Lambda^\dagger = -\Lambda$$

Geometric ViewPoint of BRST :

$$\int DX^M Dg_{\alpha\beta} e^{iS^{\text{orig}}} = Z$$

However there are redundancies in this sum as we are counting elements which are equivalent up to Weyl/Diffeos. To account for this we integrate over a space of reps for each Gauge equivalent term

→ Restrict the integral to the Quotient Space

$$(E \times M_g) / G$$

$M_g \equiv$ (Space of Metrics on Σ_g)

$E \equiv$ (Space of all embedding maps
 $X: \Sigma_g \rightarrow \mathbb{R}^D$)

$G \equiv \text{Diff}(\Sigma_g) \times \text{Weyl}(\Sigma_g)$

• The Action of G on $E \times M_g$ is defined

$$\text{by } f: \Sigma_g \rightarrow \Sigma_g : \gamma \rightarrow f^* \gamma$$

$$(f, e^f)(X, \gamma) \equiv (f^* X, e^f f^* \gamma)$$

* If we now apply this to our Path int.

$$Z = \int \mathcal{D}X \mathcal{D}g_{\alpha\beta} e^{iS^{\text{orig}}}$$

$$\int D\lambda = \text{Int. over } \mathbb{C}$$

$$\int Dg_{ab} \equiv \text{Int. over } M_g$$

Thus, the correct choice of int. over eq. classes:

$$\int [Dx][Dg] = \int_{\text{ex} M_g} D\lambda Dg / \text{Vol}(\text{Diff}(\Sigma_g) \times \text{Weyl}(\Sigma_g))$$

* We can now introduce Fad. P.P.:

$$1 = \Delta_{FP} \int d^2 \gamma \delta(\sqrt{-g} g_{\gamma}^{\alpha\beta} - \sqrt{\tilde{g}} \tilde{g}^{\alpha\beta})$$

where γ : Diffeomorphism

$$\gamma: g \mapsto g\gamma \equiv e^{\phi} \left(\partial \xi / \partial \xi' \right)^2 g(\xi)$$

$\rightarrow$ So $\gamma \in G$, i.e. this is the group action on $M_g \times \mathbb{C}$.

An aside on F.P. Ghosts & determinants

* Consider $\mathcal{M} = \mathcal{M}_{YM}[A_\mu]$

$$S[A_\mu^U] = S[A_\mu]$$

$$U(x) = \exp(-g T^a \Lambda_a(x))$$

$\rightarrow \int DU \equiv (\text{Integration over the Gauge group})$

(For us this is, of course, $G = \text{Diff}_\Sigma \times \text{Weyl}$)

* If we int. over the alg- instead, i.e.

$$\int DU = \prod_a D\Lambda_a$$

* Also, we assume invariance under

$$\int DU f(U) = \int DU f(U U')$$

i.e. it is (left) invariant under the group action.

* Consider: ✓ Functional ∂ -f

$$\Delta[A_\mu] = \int DU \delta[F_a(A_\mu^U)]$$

$$\delta[F_a] \equiv \int_a \delta[F_a]$$

- $\Delta[A_{\mu\nu}]$ is invariant under Gauge transf.'s
 $\Leftrightarrow \Delta[A_{\mu\nu}^0] = \Delta[A_{\mu\nu}]$

- We will have

$$\Delta^{-1}[H_{\mu\nu}] \Delta[A_{\mu\nu}] = I$$

With this we rewrite the above as:

$$Z = \int DA^\mu \Delta^{-1}[A_{\mu\nu}] \int DU \delta[F_a(A_{\mu\nu}^0)] e^{iS}$$

$A \rightarrow A^0$

$$\begin{aligned} \int DA^\mu e^{iS[A]} &= \int DA^\mu \Delta^{-1}[A_{\mu\nu}] \int DU \delta[F_a(A_{\mu\nu}^0)] e^{iS[A^0]} \\ &= \int DU \left[\int DA^\mu \Delta^{-1}[A_{\mu\nu}] \delta[F_a(A_{\mu\nu})] e^{iS[A]} \right] \end{aligned}$$

The DU integral has factored out of the expression:

Thus:

$$\int DU = \int \pi D\Lambda_a$$

We would like to have this integral be over Gauge fixed terms $\int DF_a$

$$\delta\Lambda_a(x) = \int d^4y \frac{\delta\Lambda_a(x)}{\delta F_b(y)} \delta F_b(y)$$

However for integrals we know we must have $\left| \frac{\partial(y^\mu)}{\partial x^\mu} \right| = \text{"det. of Jacobian"}$

$$\int DU = \int \pi_a D\Lambda_a = \int \pi_a DF_a \det \left(\frac{\delta\Lambda_a(x)}{\delta F_b(x')} \right)$$

This implies

$$\Delta[A_{\mu\nu}] = \int DU \delta F = \int \pi_a DF_a \det \left(\frac{\delta\Lambda_a}{\delta F_b} \right) \delta F$$

$\Delta[A_{\mu\nu}] = \int \pi_a DF_a \det \left(\frac{\delta\Lambda_a}{\delta F_b} \right) \delta F$

$$\Delta[A_{\mu a}] = \det \left[\frac{\delta F_a(x')}{\delta A_b(x)} \right]_{F_a=0}$$

Using $\Delta^{-1} \Delta = 1$ we have

$$\Delta^{-1}[\Delta_{\mu a}] = \det \left(\frac{\delta F_a(x')}{\delta A_b(x)} \right) \Big|_{F_a=0}$$

Back to the string:

* Looking at the above it seems rather childish. Instead let us be Geometric (See Nakamura):

On the tangent space of Σ_g : $T\Sigma_g$ we have the inner product

$$(\delta\phi_1, \delta\phi_2)_g := \int d^2\sigma \sqrt{g} g(\delta\phi_1, \delta\phi_2)$$

• We can normalize this via

$$\int D_g \delta\phi e^{-\frac{1}{2}(\delta\phi, \delta\phi)_g} = \frac{1}{\sqrt{\det(g)}}$$

$\Leftrightarrow$ This induces a measure on the field space $\int D\phi \sqrt{\det(g)}$

Scalar: $(\delta f, \delta f)_g \equiv \int d^2\xi \sqrt{g} \delta f^2$

Vector: $(\delta V^a, \delta V^b)_g \equiv \int d^2\xi \sqrt{g} g_{ab} \delta V^a \delta V^b$

Symm-tensor:

$$(\delta T_{ab}, \delta T_{ab})_g \equiv \int d^2\xi \sqrt{g} G^{abcd} T_{ab} T_{cd}$$

where $G^{abcd} \equiv G_+^{abcd} + 2g^{ab}g^{cd}$

$$G_+^{abcd} \equiv g^{ac}g^{bd} + g^{ad}g^{cb} - g^{ab}g^{cd}$$

G^{abcd} : de-Witt Metric

G_+ : Projection onto the traceless component of a tensor

ex: $g^{ab} T_{ab} = 0 \quad \Delta g_{ab} \Rightarrow g^{ab} g_{ab} \Delta = D \Delta$

$$\begin{cases} G^{abcd} T_{cd} = G^{abcd} T_{cd} = 2 T_{ab} \\ G^{abcd} \Delta g_{cd} = 2 \Delta g^{ab} \end{cases}$$

- The vector measure is $\text{Weyl}(\Sigma_g)$ invariant
- The Metric has the decomposition

$$\delta g_{ab} = \underbrace{\delta \Lambda g_{ab}}_{\text{trace}} + \underbrace{\delta g_{ab}^\perp}_{\text{traceless}}$$

$$g^{ab} \delta g_{ab} = 2 \delta \Lambda \Rightarrow \delta \Lambda = \frac{1}{2} g^{ab} \delta g_{ab}$$

$$g^{ab} \delta g_{ab} = 0$$

* We can use the above metric for δg_{ab}

$$\begin{aligned} |\delta g_{ab}|_g^2 &= (\delta g_{ab}, \delta g_{ab})_g \\ &= 4\Lambda |\delta \Lambda|_g^2 + |\delta g_{ab}^\perp|_g^2 \end{aligned}$$

→ This turns into products/sums over scalar/traceless-symm.

* This implies the measure factors as

$$Dg_{ab} = D\Lambda Dg_{ab}^\perp$$

F.P. Gauge fixing: $(f^*g, f^*\psi) \sim (g, \psi)$

$\Leftrightarrow$ Extract

$$d(\text{fields}) = |\text{Jacobian}| \times D(\text{Gauge}) \times D(\text{physical})$$

* Moduli-Space: Gauge equivalent metrics

$$\mathcal{M}_g \equiv \text{Met}(\Sigma_g) / G$$

Endow $\mathcal{M}_g$ with ^{local} coordinates t_i

$t_i \equiv$ Model parameters, $i = (1, \dots, \dim(\mathcal{M}_g))$

Teichmüller Space: $\mathcal{T}_g \equiv \text{Met}(\Sigma_g)/G_0$
 $G_0 = \text{GL}(n, \mathbb{R}) \rtimes \mathbb{R}^n$

* Modular group: $\Gamma_g \equiv \pi_0(\text{Diff}(\Sigma_g)) = \frac{\text{Diff}(\Sigma)}{\text{Diff}_0(\Sigma)}$

$$\mathcal{M}_g = \mathcal{T}_g / \Gamma_g$$

$$\mathcal{M}_g \equiv \dim_{\mathbb{R}}(\mathcal{M}_g) = \dim_{\mathbb{R}}(\mathcal{T}_g)$$

Their volumes are related by

$$\int_{\mathcal{M}_g} D^{\mathcal{M}_g} t = \frac{1}{\Omega_{\mathcal{T}_g}} \int_{\mathcal{T}_g} d^{\mathcal{M}_g} t$$

Def: Let G be a group, then the Haar Measure (dg) , which is left-invariant, can be integrated over the entire group G to yield the volume:

$$\Omega_G \equiv \int_G dg = \int d(hg) \quad \forall h \in G$$

Now, given the Lie Alg. of G , $\mathfrak{g}$, with generators T_i we have

$$\alpha = \alpha^i T_i \in \mathfrak{g}$$

Thus, we can parametrize the integral using the coefficients α^i

$$\Omega_G = \int d\alpha = \int \prod_i d\alpha^i$$

Claim: $\mathcal{T}_g$ is a mfd while $\mathcal{M}_g$ is NOT for $g \geq 2$. Here $\mathcal{M}_g$ is an orbifold.

* Now

$$g_{ab} = \hat{g}_{ab}(t) = e^{2f^*} \hat{g}_{ab}(t) = f^*(e^{2f} \hat{g}_{ab})$$

$$g_{ab}|_t = \tilde{g}_{ab}^{(f,d)}(a,t)$$

- Now, we are going from
 $g \rightarrow [g]$

The space M_g has coordinates t_i and so let us Recall:

- Take $x^M \rightarrow x^M + \delta x^M \in M$

then δx^M is a tangent vector on TM which is subject to $g: TM \times TM \rightarrow K$

Now, M_g has coordinates t_i (which represent $[\tilde{g}_{ab}] = t_i$ equivalence classes).

- ① Thus, $t_i \rightarrow t_i + \delta t_i$ then

$\delta t_i \in TM_g$ is a tangent space element, which is subject to a Differential metric.

- ② Note, however, what δt_i means: With δt_i , just as with δx^M , we are essentially trying to understand how the metric we are looking at/using is going to change/compare to those in different eq. classes.

As such, to get an idea of what to look at "locally" we have

$$t_i = t_i([g_{ab}])$$

$$\therefore \delta t_i = \frac{\partial t_i}{\partial g^{ab}} \delta g^{ab}$$

(terchmüller)
Deformations

Q: How do I know I can invert the relationship between t_i and g ?
 i.e. how do I know $\partial t_i / \partial g^{ab}$ is invertible.

I'm sure the answer is straight forward
 but I don't see it.

▷ Inverting this relationship we find

$$\delta_t g_{ab} = \frac{\delta g_{ab}[t_i]}{\partial t_i} \delta t_i$$

This term must be added to the overall variation of g :

$$\delta g_{ab} = 2\phi g_{ab} + \nabla_a V_b + \nabla_b V_a + \delta t_i \partial_i g_{ab}$$

* Since we know we can integrate over $\text{Diff}(M)$ and $\text{Weyl}(M)$:

$$\Omega_{\text{Diff}} := \int Dg V \quad \text{and} \quad \Omega_{\text{Weyl}} := \int Dg \phi$$

where

$$\Omega_{\text{Diff}}^{[g]} = \Omega_{\text{Diff}}[g] \Omega_{\Gamma_g}$$

Claim: Ω_{Diff} is invariant under Weyl :

Proof:

$$\Omega_{\text{Diff}}[e^{2\phi} \hat{g}] = \int D e^{2\phi} \hat{g} V = \int D e^{2\phi} \hat{g} V = \Omega[\hat{g}]$$

Gauge Fixing term: $F_{ab} = g_{ab} - g_{ab}^{(f,d)}(t) = 0$
 $\gamma \equiv (f, \phi)$

* We factor δg_{ab} (traceless) + (trace)

$$\begin{aligned} g^{ab} \delta g_{ab} &= 4\phi + 2g^{ab} \nabla_a V_b + \delta t_i g^{ab} \partial_i g_{ab} \\ &= 4\phi + 2\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab} \\ &= 4\Lambda \end{aligned}$$

Since the traceless term vanishes:

$$g^{ab} \delta g_{ab} \Leftrightarrow$$

$$\hat{\Lambda} = \phi + \frac{1}{2} (\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab})$$

$$\bullet \text{ Define } \Lambda = \phi + \frac{1}{2} \nabla_a V^a$$

$$\tilde{\Lambda} = \Lambda + \frac{1}{2} \delta t_i g^{ab} \partial_i g_{ab}$$

* Plug ϕ back into the expression
 $\phi \equiv \tilde{\Lambda} - \frac{1}{2} (\nabla_a V^a + \delta t_i g^{ab} \partial_i g_{ab})$

$$\begin{aligned} \delta g_{ab} &= 2 \left[\tilde{\Lambda} - \frac{1}{2} (\nabla_a V^a + \delta t_i g^{cd} \partial_i g_{cd}) \right] g_{ab} \\ &\quad + \nabla_a V_b + \nabla_b V_a + \delta t_i \partial_i g_{ab} \\ &= 2 \tilde{\Lambda} g_{ab} + (\nabla_a V_b + \nabla_b V_a - g_{ab} \nabla_c V^c) \\ &\quad - \delta t_i (\partial_i g_{ab} - g_{ab} g^{cd} \partial_i g_{cd}) \end{aligned}$$

• traceless: $2 \tilde{\Lambda} g_{ab}$

• Diffeomorphism: $\nabla_a V_b + \nabla_b V_a - g_{ab} \nabla_c V^c \equiv (P_i V)_{ab}$

• Beltrami: $M_{iab} \equiv \partial_i g_{ab} - g_{ab} g^{cd} \partial_i g_{cd}$

* We simply found the trace and then grouped together the terms involving V^a and δt_i .

$$Z_g = \int D^{\mu g} t. D_g \tilde{\Lambda} D_g (P_i V) \Omega_{\text{gauge}} [g]^{-1} Z_m [g]$$

To use this Measure we must relate it to the Measure we have established over (V, ϕ)

• Change of coordinates:

$$(P_i V, \tilde{\Lambda}) \longrightarrow (V, \phi)$$

$$D_g (P_i V) D_g \tilde{\Lambda} = D_g V D_g \phi \Delta_{FP} [g]$$

The FP-Det. is simply the Jacobian

$$\Delta_{FP} [g] = \det \left(\frac{\partial (P_i V, \tilde{\Lambda})}{\partial (V, \phi)} \right)$$

Formally, we write this as

$$\partial P_i V / \partial \phi = 0$$

$$\Delta_{FP}[g] = \begin{vmatrix} P_1 & 0 \\ \partial \tilde{\Lambda} / \partial V & 1 \end{vmatrix}$$

$$\Delta_{FP}[g] = \det(P_1)$$

* This is a very rough argument. A more rigorous analysis: Nakahara

Back to West: (3.2.20)

After all that, we find that the above determinant is equivalent to Eq. 3.2.20

Since

$$\Delta_{FP} \det(P_1) = \det \left(\sqrt{-g} \frac{\delta}{\delta \zeta^\alpha} (P_1 \zeta)^\alpha \right) \Big|_{\zeta^\alpha = 0}$$

$$\text{or, } \det(P_1) = \Delta_{FP} = \det \left(\frac{\delta}{\delta \zeta^\alpha} (\sqrt{-g} g^{\alpha\beta} \zeta_\beta) \right)$$

$$\Delta_{FP} = \det \left\{ \sqrt{-g} (\nabla^\alpha \delta_\gamma^\beta + \nabla^\beta \delta_\gamma^\alpha - g^{\alpha\beta} \nabla_\gamma) \delta^2(\zeta - \zeta') \right\}$$

$$\text{Since } \delta \zeta^\alpha(\zeta) / \delta \zeta^\beta(\zeta') = \delta^\alpha_\beta \delta^2(\zeta - \zeta')$$

$$P_1: TM \rightarrow (Sym^2 TM)|_{\text{traceless}}$$

* We have, via the inner products defined above, the adjoint

$$P_1^\dagger: (Sym^2 TM)|_{\text{traceless}} \rightarrow TM$$

$$((P_1 V)_{\alpha\beta}, t_{\alpha\beta})_2 = (V_\alpha, (P_1^\dagger t)_\alpha)_1$$

where ${}_2$: metric on $Sym^2 TM$

${}_1$: metric on TM

metric on TM

$$(P^\dagger t)_a = -2 \nabla^b t_{ab}$$

(Adjoint Map)

* Now, Recall that we have found the Action for the Ghost field and Gauge fixed term:

$$S^{gh} = -\frac{1}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (\nabla^\alpha c^\beta + \nabla^\beta c^\alpha - g^{\alpha\beta} \nabla_c c^c)$$

$$\text{or, } S^{gh} = -\frac{1}{\pi} \int d^2 \xi \sqrt{-g} b_{\alpha\beta} (P_1 c)^{\alpha\beta}$$